

SKILL WEEK12

TASK1: To execute large Linux pipelines, run multiple jobs, monitor CPU/memory usage, detect failures, and analyze execution time.

```
#!/bin/bash

echo "Starting performance test..."

time seq 1 1000000 | grep "999" | sort | uniq | wc -l

if [ $? -eq 0 ]; then
    echo "Pipeline completed successfully"
else
    echo "Pipeline failed"
fi

echo "Performance test completed."
```

```
laasya@laasya:~/pipeline_lab$ sleep 20 &
sleep 30 &
sleep 40 &
[1] 1134
[2] 1135
[3] 1136
```

```
laasya@laasya:~/pipeline_lab$ jobs
[1]    Running                sleep 20 &
[2]-  Running                sleep 30 &
[3]+  Running                sleep 40 &
```

```
laasya@laasya:~/pipeline_lab$ ps
```

```
  PID TTY          TIME CMD
 1101 pts/2    00:00:00 bash
 1134 pts/2    00:00:00 sleep
 1135 pts/2    00:00:00 sleep
 1136 pts/2    00:00:00 sleep
 1137 pts/2    00:00:00 ps
```

```
laasya@laasya:~/pipeline_lab$ top
```

```
top - 15:19:15 up 1:09, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 23 total, 1 running, 22 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.
MiB Mem :  7762.6 total,  7154.6 free,   532.3 used,   226.7 buff/cache
MiB Swap:  2048.0 total,  2048.0 free,    0.0 used.  7230.3 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+
1	root	20	0	21736	13036	9676	S	0.0	0.2	0:00.66
2	root	20	0	3180	2204	2072	S	0.0	0.0	0:00.01
7	root	20	0	3196	2128	2008	S	0.0	0.0	0:00.00
53	root	19	-1	50376	16176	15076	S	0.0	0.2	0:00.27
100	root	20	0	25288	6676	5156	S	0.0	0.1	0:00.92
164	systemd+	20	0	21348	13180	11052	S	0.0	0.2	0:00.07
170	systemd+	20	0	91040	7944	6972	S	0.0	0.1	0:00.20
176	root	20	0	4248	2676	2428	S	0.0	0.0	0:00.02
177	message+	20	0	9648	5420	4688	S	0.0	0.1	0:00.09
184	root	20	0	17980	8784	7744	S	0.0	0.1	0:00.09
194	syslog	20	0	222520	5652	4636	S	0.0	0.1	0:00.08
197	root	20	0	3128	1972	1832	S	0.0	0.0	0:00.00
208	root	20	0	107040	23284	13828	S	0.0	0.3	0:00.10
298	root	20	0	6704	4652	3872	S	0.0	0.1	0:00.00
341	laasya	20	0	20336	11476	9352	S	0.0	0.1	0:00.05

```

[1] Done sleep 20
[2]- Done sleep 30
[3]+ Done sleep 40
laasya@laasya:~/pipeline_lab$ ls existing_file.txt
echo $?
ls: cannot access 'existing_file.txt': No such file or directory
2
laasya@laasya:~/pipeline_lab$ ls
echo $?
0
laasya@laasya:~/pipeline_lab$ echo $?
0
laasya@laasya:~/pipeline_lab$ nano performance.sh
laasya@laasya:~/pipeline_lab$ chmod +x performance.sh
laasya@laasya:~/pipeline_lab$ ./performance.sh
Starting performance test...
3700

real    0m0.015s
user    0m0.016s
sys     0m0.014s
Pipeline completed successfully
Performance test completed.
laasya@laasya:~/pipeline_lab$ |

```

TASK 2: Document Architecture, Create Flow Diagrams, Explain Execution Paths, Describe Syscalls, Record Design Decisions, Review Documentation

The major system calls involved are:

```

fork()
pipe()
dup2()
exec()
wait()
close()

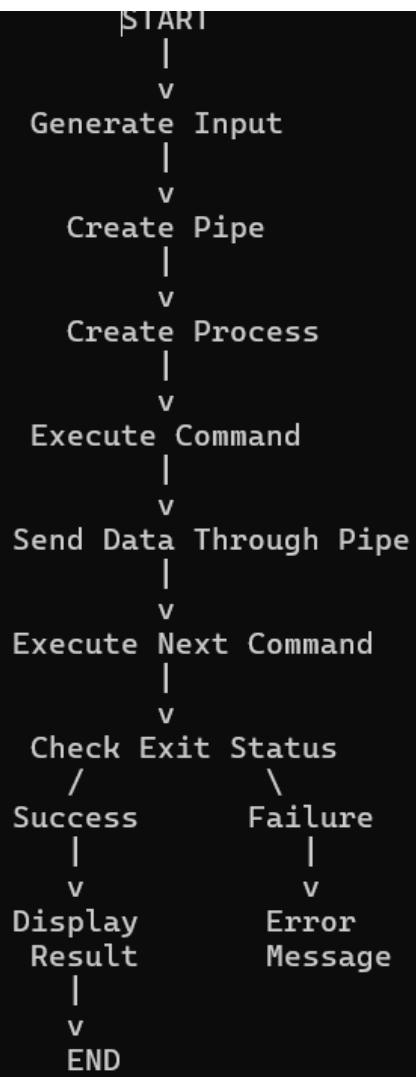
```

Design Decisions

- Pipes are used for communication between processes.
- Separate processes execute individual commands.
- Exit status is checked to detect failures.
- Performance is measured using the time command.

Testing

The pipeline is tested with large input values and multiple jobs.



```
laasya@laasya:~/pipeline_lab$ mkdir documentation
laasya@laasya:~/pipeline_lab$ cd documentation
laasya@laasya:~/pipeline_lab/documentation$ nano README.md
laasya@laasya:~/pipeline_lab/documentation$ nano flow.txt
laasya@laasya:~/pipeline_lab/documentation$ sudo apt update
[sudo] password for laasya:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
107 packages can be upgraded. Run 'apt list --upgradable' to see them.
laasya@laasya:~/pipeline_lab/documentation$ sudo apt install strace
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libdrm-nouveau2 libdrm-radeon1 libgl1-amber-dri libglapi-mesa
  libllvm17t64 libwayland-server0 libxcb-dri2-0
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  strace
0 upgraded, 1 newly installed, 0 to remove and 107 not upgraded.
Need to get 584 kB of archives.
After this operation, 2153 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 strace amd64 6.8-0ubuntu2 [584 kB]
Fetched 584 kB in 1s (390 kB/s)
Selecting previously unselected package strace.
(Reading database ... 47884 files and directories currently installed.)
Preparing to unpack .../strace_6.8-0ubuntu2_amd64.deb ...
Unpacking strace (6.8-0ubuntu2) ...
```

```

[pid 1576] fstat(3, {st_mode=S_IFREG|0644, st_size=3360, ...}) = 0
[pid 1576] mmap(NULL, 3360, PROT_READ, MAP_PRIVATE, 3, 0) = 0x74e3d9128000
[pid 1576] close(3) = 0
[pid 1576] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_NUMERIC", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
[pid 1576] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_NUMERIC", O_RDONLY|O_CLOEXEC) = 3
[pid 1576] fstat(3, {st_mode=S_IFREG|0644, st_size=50, ...}) = 0
[pid 1576] mmap(NULL, 50, PROT_READ, MAP_PRIVATE, 3, 0) = 0x74e3d9127000
[pid 1576] close(3) = 0
[pid 1576] openat(AT_FDCWD, "/usr/lib/locale/C.UTF-8/LC_CTYPE", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
[pid 1576] openat(AT_FDCWD, "/usr/lib/locale/C.utf8/LC_CTYPE", O_RDONLY|O_CLOEXEC) = 3
[pid 1576] fstat(3, {st_mode=S_IFREG|0644, st_size=360460, ...}) = 0
[pid 1576] mmap(NULL, 360460, PROT_READ, MAP_PRIVATE, 3, 0) = 0x74e3d90ce000
[pid 1576] close(3) = 0
[pid 1576] fstat(0, {st_mode=S_IFIFO|0600, st_size=0, ...}) = 0
[pid 1576] fadvise64(0, 0, 0, POSIX_FADV_SEQUENTIAL) = -1 ESPIPE (Illegal seek)
[pid 1576] read(0, "hello\n", 16384) = 6
[pid 1576] read(0, "", 16384) = 0
[pid 1576] fstat(1, {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0x2), ...}) = 0
[pid 1576] write(1, "6\n", 26) = 2
[pid 1576] close(0) = 0
[pid 1576] close(1) = 0
[pid 1576] close(2) = 0
[pid 1576] exit_group(0) = ?
[pid 1576] +++ exited with 0 +++
<... wait4 resumed>[{WIFEXITED(s) && WEXITSTATUS(s) == 0}], 0, NULL) = 1576
--- SIGCHLD {si_signo=SIGCHLD, si_code=CLD_EXITED, si_pid=1576, si_uid=1000, si_status=0, si_etime=0, si_stime=2 /* 0.02 s */} ---
rt_sigreturn({mask=[]}) = 1576
wait4(-1, 0x7ffea92a452c, WNOHANG, NULL) = -1 ECHILD (No child processes)
exit_group(0) = ?
+++ exited with 0 +++
laasya@laasya:~/pipeline_lab/documentation$ |

```

TASK 3: Organize Source Code, Improve Build Scripts, Add README, Verify Repository Structure, Tag Releases, Validate Deliverables

```

#ifndef CALCULATOR_H
#define CALCULATOR_H

int add(int a, int b);
int subtract(int a, int b);

#endif

```

```
#include "calculator.h"

int add(int a, int b)
{
    return a + b;
}

int subtract(int a, int b)
{
    return a - b;
}
```

```
#include <stdio.h>
#include "calculator.h"

int main()
{
    int a = 20;
    int b = 10;

    printf("Addition: %d\n", add(a, b));
    printf("Subtraction: %d\n", subtract(a, b));

    return 0;
}
```

```
CC = gcc
CFLAGS = -Wall -Iinclude

TARGET = calculator

SRC = src/main.c src/calculator.c

all:
    $(CC) $(CFLAGS) $(SRC) -o $(TARGET)

clean:
    rm -f $(TARGET)

run:
    ./$(TARGET)
```

```
# Linux Calculator Project

## Description
This project demonstrates modular C programming and automated building using Makefile.

## Structure
include/
    calculator.h
src/
    main.c
    calculator.c
tests/
    test.sh

Makefile
README.md

## Build
make

## Run
./calculator

## Clean
make clean

## Output
The program performs addition and subtraction.
```

```
Processing triggers for man-db (2.12.0-4build2) ...
laasya@laasya:~/pipeline_lab/documentation/final_project$ nano include/calculator.h
laasya@laasya:~/pipeline_lab/documentation/final_project$ nano src/calculator.c
laasya@laasya:~/pipeline_lab/documentation/final_project$ nano src/main.c
laasya@laasya:~/pipeline_lab/documentation/final_project$ nano Makefile
laasya@laasya:~/pipeline_lab/documentation/final_project$ make
gcc -Wall -Iinclude src/main.c src/calculator.c -o calculator
laasya@laasya:~/pipeline_lab/documentation/final_project$ ./calculator
Addition: 30
Subtraction: 10
laasya@laasya:~/pipeline_lab/documentation/final_project$ make run
./calculator
Addition: 30
Subtraction: 10
laasya@laasya:~/pipeline_lab/documentation/final_project$ make run
./calculator
Addition: 30
Subtraction: 10
laasya@laasya:~/pipeline_lab/documentation/final_project$ make clean
rm -f calculator
laasya@laasya:~/pipeline_lab/documentation/final_project$ ls
Makefile include src tests
laasya@laasya:~/pipeline_lab/documentation/final_project$ make
gcc -Wall -Iinclude src/main.c src/calculator.c -o calculator
laasya@laasya:~/pipeline_lab/documentation/final_project$ nano README.md
laasya@laasya:~/pipeline_lab/documentation/final_project$ git init
hint: Using 'master' as the name for the initial branch. This default branch
name
hint: is subject to change. To configure the initial branch name to use in a
ll
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
```

```

nothing added to commit but untracked files present (use "git add" to track)
laasya@laasya:~/pipeline_lab/documentation/final_project$ git add .
laasya@laasya:~/pipeline_lab/documentation/final_project$ git commit -m "Initial project release"
[master (root-commit) cf14c11] Initial project release
 6 files changed, 82 insertions(+)
 create mode 100644 Makefile
 create mode 100644 README.md
 create mode 100755 calculator
 create mode 100644 include/calculator.h
 create mode 100644 src/calculator.c
 create mode 100644 src/main.c
laasya@laasya:~/pipeline_lab/documentation/final_project$ git commit -m "Initial project release"
On branch master
nothing to commit, working tree clean
laasya@laasya:~/pipeline_lab/documentation/final_project$ git tag v1.0
laasya@laasya:~/pipeline_lab/documentation/final_project$ git tag v1.0
laasya@laasya:~/pipeline_lab/documentation/final_project$ tree
.
├── Makefile
├── README.md
├── calculator
├── include
│   └── calculator.h
├── src
│   ├── calculator.c
│   └── main.c
└── tests

4 directories, 6 files
laasya@laasya:~/pipeline_lab/documentation/final_project$

```

TASK 4: Create Demo Scenarios, Validate Features, Rehearse Presentation, Demonstrate Pipelines, Demonstrate Signals, Final Project Review

```

laasya@laasya:~/pipeline_lab/documentation/final_project$ git commit -m "Initial project release"
On branch master
nothing to commit, working tree clean
laasya@laasya:~/pipeline_lab/documentation/final_project$ git tag v1.0
laasya@laasya:~/pipeline_lab/documentation/final_project$ git tag
v1.0
laasya@laasya:~/pipeline_lab/documentation/final_project$ tree
.
├── Makefile
├── README.md
├── calculator
├── include
│   └── calculator.h
├── src
│   ├── calculator.c
│   └── main.c
└── tests

4 directories, 6 files
laasya@laasya:~/pipeline_lab/documentation/final_project$ mkdir demo
laasya@laasya:~/pipeline_lab/documentation/final_project$ cd demo
laasya@laasya:~/pipeline_lab/documentation/final_project/demo$ seq 1 10000 |
grep "99" | wc -l
280
laasya@laasya:~/pipeline_lab/documentation/final_project/demo$ sleep 30 &
sleep 40 &
sleep 50 &
[1] 1788
[2] 1789
[3] 1790
laasya@laasya:~/pipeline_lab/documentation/final_project/demo$ jobs
[1]  Running                  sleep 30 &
[2]-  Running                  sleep 40 &
[3]+  Running                  sleep 50 &

```

```
laasya@laasya:~/pipeline_lab/documentation/final_project/demo$ top
top - 15:43:26 up 1:34, 1 user, load average: 0.10, 0.04, 0.01
Tasks: 23 total, 1 running, 22 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.
MiB Mem : 7762.6 total, 6716.6 free, 536.9 used, 663.3 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used. 7225.7 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+
1799	laasya	20	0	9304	5684	3496	R	0.3	0.1	0:00.03
1	root	20	0	21888	13060	9676	S	0.0	0.2	0:00.75
2	root	20	0	3180	2204	2072	S	0.0	0.0	0:00.01
7	root	20	0	3196	2128	2008	S	0.0	0.0	0:00.00
53	root	19	-1	50376	16208	15096	S	0.0	0.2	0:00.37
100	root	20	0	25288	6676	5156	S	0.0	0.1	0:01.24
164	systemd+	20	0	21348	13180	11052	S	0.0	0.2	0:00.08
170	systemd+	20	0	91040	7944	6972	S	0.0	0.1	0:00.27
176	root	20	0	4248	2676	2428	S	0.0	0.0	0:00.02
177	message+	20	0	9648	5424	4688	S	0.0	0.1	0:00.12
184	root	20	0	17980	8784	7744	S	0.0	0.1	0:00.10
194	syslog	20	0	222520	5652	4636	S	0.0	0.1	0:00.09
197	root	20	0	3128	1972	1832	S	0.0	0.0	0:00.00
208	root	20	0	107040	23284	13828	S	0.0	0.3	0:00.10
298	root	20	0	6704	4652	3872	S	0.0	0.1	0:00.00
341	laasya	20	0	20336	11476	9352	S	0.0	0.1	0:00.05
342	laasya	20	0	21176	3596	1868	S	0.0	0.0	0:00.00
366	laasya	20	0	6084	5388	3704	S	0.0	0.1	0:00.01
424	laasya	20	0	2692	876	776	S	0.0	0.0	0:00.29
911	polkitd	20	0	308176	8120	7224	S	0.0	0.1	0:00.20
1093	root	20	0	3188	1116	980	S	0.0	0.0	0:00.00
1094	root	20	0	3204	1228	1080	S	0.0	0.0	0:00.23
1101	laasya	20	0	6728	5972	3692	S	0.0	0.1	0:00.21

```
#include <stdio.h>
#include <signal.h>
#include <unistd.h>

void handle_signal(int sig)
{
    printf("\nSIGINT received. Program is still running.\n");
}

int main()
{
    signal(SIGINT, handle_signal);

    printf("Program started. Press Ctrl+C to send SIGINT.\n");

    while (1)
    {
        printf("Running...\n");
        sleep(2);
    }

    return 0;
}
```

```

[1] Done sleep 30
[2]- Done sleep 40
[3]+ Done sleep 50
laasya@laasya:~/pipeline_lab/documentation/final_project/demo$ time sh -c 'seq 1 1000000 | grep "999" | sort | uniq | wc -l'
3700

real    0m0.051s
user    0m0.058s
sys     0m0.046s
laasya@laasya:~/pipeline_lab/documentation/final_project/demo$ nano signal_demo.c
laasya@laasya:~/pipeline_lab/documentation/final_project/demo$ gcc signal_demo.c -o signal_demo
laasya@laasya:~/pipeline_lab/documentation/final_project/demo$ ./signal_demo
Program started. Press Ctrl+C to send SIGINT.
Running...
Running...
^C
SIGINT received. Program is still running.
Running...
Running...
Running...
Running...
^Z
[1]+ Stopped ./signal_demo
laasya@laasya:~/pipeline_lab/documentation/final_project/demo$ |

```

```

laasya@laasya:~/pipeline_lab/documentation/final_project/demo$ cd ~/final_project
-bash: cd: /home/laasya/final_project: No such file or directory
laasya@laasya:~/pipeline_lab/documentation/final_project/demo$ ./calculator
-bash: ./calculator: No such file or directory
laasya@laasya:~/pipeline_lab/documentation/final_project/demo$ |

```